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Consider sorting three numbers a_1 , a_2 , and a_3 with comparison-based sorting algorithms then how many combinations are possible?

Answer ([Please click here to Add Answer](#))





How edit distance is used for speech reorganization?

Answer ([Please click here to Add Answer](#))



What will be the sorting techniques by analyzing the followings?

- (i) Whenever two consecutive items are found in an array that is out of order, swap them.
- (ii) Assume that $A[1 \dots i - 1]$ have already been sorted. Add $A[i]$ into its proper position in this sub array. Create this position by shifting all larger elements to the right.
- (iii) Assume that $A[1 \dots i - 1]$ contain the $i - 1$ smallest elements in sorted order. Find the smallest element in $A[i \dots n]$ Swap it with $A[i]$.

Answer ([Please click here to Add Answer](#))



True or False: A sequence of values in a row of the dynamic programming table for an instance of the knapsack problem is always non-decreasing. **Give a brief description.**

Answer ([Please click here to Add Answer](#))



Draw the final **Max-Heap** structure for the following array,

50, 31, 45, 30, 2, 7, 40, 12, 28, 1

You can show the final result (tree) only.

Answer (Please [click here](#) to Add Answer)



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Describe three solution paths for edit distance of these two given words,

1. MATHS
2. ARTS

Answer ([Please click here to Add Answer](#))



What is the essential constraint for the Counting Sort?

Answer ([Please click here to Add Answer](#))



How we proceed with m entries in cost table for chain matrix multiplication problem?

Answer ([Please click here to Add Answer](#))



Solve it,

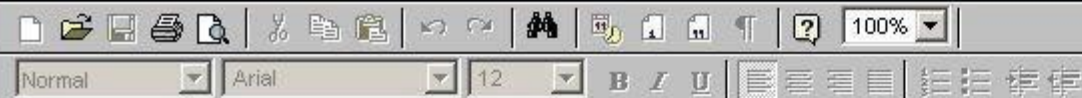
$$T(n) = \frac{1}{2} \sum_{q=1}^2 (T(q-1) + T(2-q) + 2)$$

Answer ([Please click here to Add Answer](#))



True or False: A sequence of values in a row of the dynamic programming table for an instance of the knapsack problem is always non-decreasing. **Give a brief description.**

Answer ([Please click here](#) to Add Answer)



Whether each of the following sorting algorithms is stable and in-place or not?

Sorting Algorithm	Stable	In-Place
Merge Sort	Yes/No	Yes/No
Heapsort	Yes/No	Yes/No
Quicksort	Yes/No	Yes/No
Counting Sort	Yes/No	Yes/No

Answer ([Please click here to Add Answer](#))



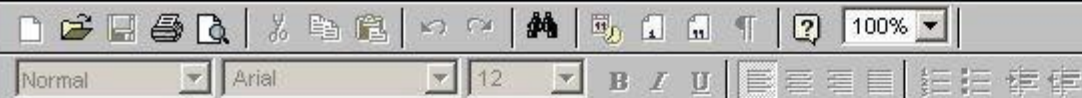
Merge Sort	Yes/No	Yes/No
Heapsort	Yes/No	Yes/No
Quicksort	Yes/No	Yes/No
Counting Sort	Yes/No	Yes/No
Bubble Sort	Yes/No	Yes/No

Answer ([Please click here to Add Answer](#))



Write down the steps of dynamic programming strategy.

Answer ([Please click here to Add Answer](#))



What is the necessary assumption for average case analysis of quick sort?

Answer ([Please click here](#) to Add Answer)



If we solve the knap-sap algorithm with brute force then how much running time will be required?

Answer ([Please click here to Add Answer](#))



In comparison based sorting algorithms, there are " $n!$ " Permutations to arrange the elements, and lower bound achieved is $\Omega(n \log n)$. To prove this, which factors of binary tree enable us to take the " $\log$ " for " $n!$ " comparisons?

Answer ([Please click here to Add Answer](#))



What are Catalan numbers? Give the formula to find Catalan numbers.

Answer ([Please click here to Add Answer](#))



Consider the following recursive `search` function which returns the index of the array element containing `key`, if such an element exists.

```
int search( int* array, int left, int right, int key )
{
    int mid = (left+right)/2;
```

Answer (Please [click here](#) to Add Answer)



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```
int mid = (left+right)/2;  
if( left == right )  
    return left;  
else if( array[mid] <= key )  
    return search( array, mid+1, right, key );
```

Answer ([Please click here to Add Answer](#))



```
else if( array[mid] <= key )  
    return search( array, mid+1, right, key );  
else  
    return search( array, left, mid, key );  
}
```

Answer ([Please click here to Add Answer](#))



```
return search( array, left, mid, key );  
}
```

Given the following declaration:

```
int list[4] = { 5, 6, 7, 8 };
```

does `search(list, 0, 4, 8)` correctly return the index of the `list` element containing 8? Show your work.

Answer ([Please click here to Add Answer](#))



Write down the steps of dynamic programming strategy.

Answer ([Please click here](#) to Add Answer)



Question No : 21 of 26

Marks: 2 (Budgeted Time 4 Min) 

What is the idea behind in counting sort algorithm, to sort the elements without comparisons in linear time?

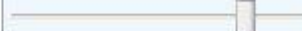
Answer ([Please click here to Add Answer](#)) 

The image shows a rich text editor interface with a toolbar at the top containing icons for file operations (new, open, save, print, find, replace, undo, redo), text formatting (bold, italic, underline, text color, background color), and alignment (left, center, right, justified). Below the toolbar is a text area where the answer can be typed. The text area is currently empty.

Start Time: 5:37 AM

60:00

Time Left

21 

Question No : 22 of 26

Marks: 2 (Budgeted Time 4 Min) 

Which one is the better approach to cop with chain matrix multiplication?

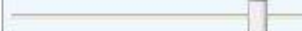
Answer ([Please click here to Add Answer](#)) 

A rich text editor toolbar with various icons for text formatting (bold, italic, underline, text color, background color), alignment (left, center, right, justified), and other functions. The font is set to Arial, size 12, and the zoom level is 100%.

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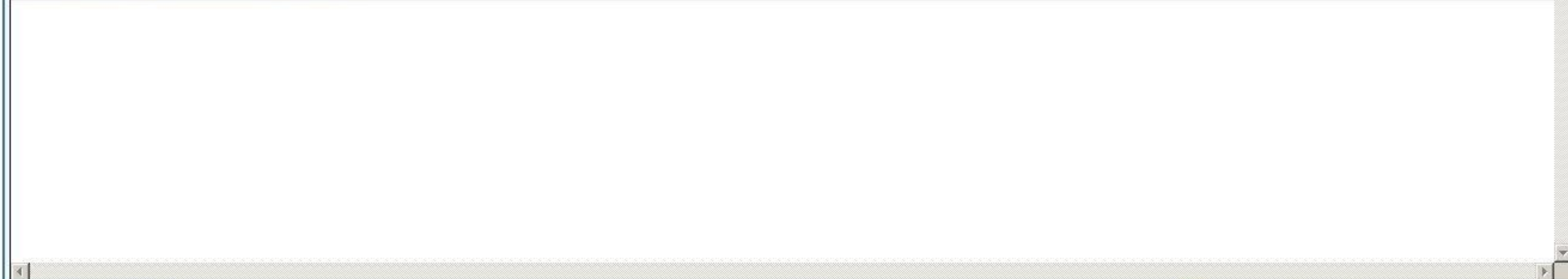
Question No : 23 of 26

Marks: 3 (Budgeted Time 6 Min) 

Consider sorting three numbers a1, a2 and a3 with comparison-based sorting algorithms then write the possible combinations?

Answer ([Please click here to Add Answer](#)) 

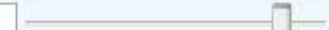
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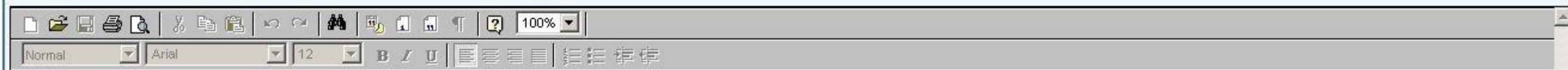
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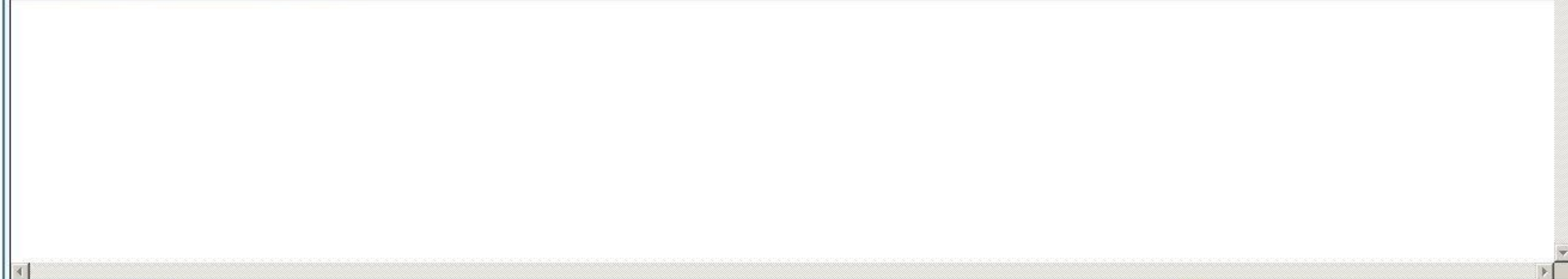
Question No : 24 of 26

Marks: 3 (Budgeted Time 6 Min) 

What are Catalan numbers? Give the formula to find Catalan numbers.

Answer ([Please click here to Add Answer](#)) 

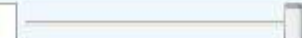
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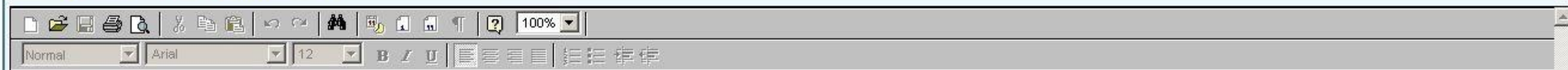
24 

Question No : 25 of 26

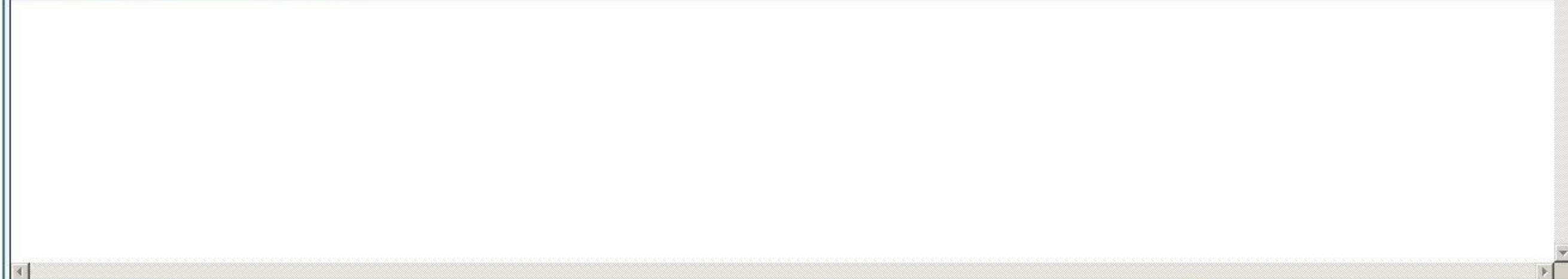
Marks: 5 (Budgeted Time 10 Min) 

The following summation form represents quick sort, so what will be the outcome after solving this recurrence in terms of asymptotic notation?

$$\sum_{i=1}^{n+1} i = \frac{(n+1)(n+2)}{2}$$

Answer (Please [click here](#) to Add Answer) 

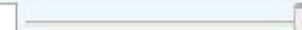
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Question No : 26 of 26

Marks: 5 (Budgeted Time 10 Min) 

What is the time complexity of the following algorithm?

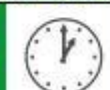
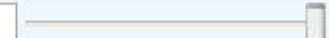
KNAPSACK(n, W)

```
1 for  $w = 0, W$ 
2 do  $V[0, w] \leftarrow 0$ 
3 for  $i = 0, n$ 
4 do  $V[i, 0] \leftarrow 0$ 
5 for  $w = 0, W$ 
```

Answer ([Please click here to Add Answer](#)) 

Rich text editor toolbar with icons for Bold, Italic, Underline, Text Color, Background Color, Bulleted List, Numbered List, Indent, Outdent, Link, Unlink, and others. The font is set to Arial, size 12, and the zoom level is 100%.

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Question No : 26 of 26

Marks: 5 (Budgeted Time 10 Min) 

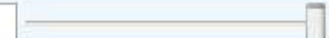
```
2 do  $v[i, w] \leftarrow v$ 
3 for  $i = 0, n$ 
4 do  $V[i, 0] \leftarrow 0$ 
5   for  $w = 0, W$ 
6     do if  $(w_i \leq w \ \& \ v_i + V[i-1, w-w_i] > V[i-1, w])$ 
7       then  $V[i, w] \leftarrow v_i + V[i-1, w-w_i]$ 
8     else  $V[i, w] \leftarrow V[i-1, w]$ 
```

Answer ([Please click here to Add Answer](#)) 

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Question No : 26 of 26

Marks: 5 (Budgeted Time 10 Min) 

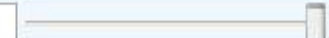
```
4  do  $V[i, w] \leftarrow 0$ 
5  for  $w = 0, W$ 
6  do if  $(w_i \leq w \ \& \ v_i + V[i-1, w-w_i] > V[i-1, w])$ 
7      then  $V[i, w] \leftarrow v_i + V[i-1, w-w_i]$ 
8      else  $V[i, w] \leftarrow V[i-1, w]$ 
```

Answer (Please [click here](#) to Add Answer) 

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Question No : 21 of 26

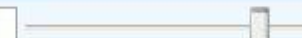
Marks: 2 (Budgeted Time 4 Min) 

What is the idea behind in counting sort algorithm, to sort the elements without comparisons in linear time?

Answer ([Please click here to Add Answer](#)) 

Start Time: 5:51 AM

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Time Left 21 

Question No : 22 of 26

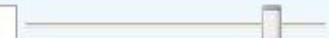
Marks: 2 (Budgeted Time 4 Min) 

How edit distance is used for speech reorganization?

Answer (Please [click here](#) to Add Answer) 

A rich text editor toolbar with various icons for text formatting and editing. The toolbar includes icons for bold, italic, underline, strikethrough, text color, background color, bulleted list, numbered list, indent, outdent, link, unlink, and a help icon. The font face is set to Arial and the font size is 12. The text color is black and the background color is white. The editor area is currently empty.

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Question No : 23 of 26

Marks: 3 (Budgeted Time 6 Min)



How to construct an Optimal Solution for 0/1 knapsack problem?

Answer (Please [click here](#) to Add Answer)

Start Time: 5:51 AM

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23



Question No : 24 of 26

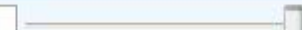
Marks: 3 (Budgeted Time 6 Min) 

In comparison based sorting algorithms, there are " $n!$ " Permutations to arrange the elements, and lower bound achieved is $\Omega(n \log n)$. To prove this, which factors of binary tree enable us to take the "log" for " $n!$ " comparisons?

Answer (Please [click here](#) to Add Answer) 

A rich text editor toolbar with various icons for text formatting (bold, italic, underline, text color, background color), alignment (left, center, right, justified), and other editing functions. The font is set to Arial, size 12. The editor area is currently empty.

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Question No : 25 of 26

Marks: 5 (Budgeted Time 10 Min) 

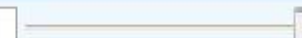
The table below shows the contents of an unsorted array. Perform radix sort on this array, show the contents of the array after the first and second pass of the sort.

Unsorted Data	After Pass 1	After Pass 2
342		
103		
...		

Answer ([Please click here to Add Answer](#)) 

Rich text editor toolbar with icons for file operations, text formatting, and alignment. The toolbar includes buttons for New, Open, Save, Print, Find, Cut, Copy, Paste, Undo, Redo, Bold, Italic, Underline, Bulleted List, Numbered List, Indent Left, Indent Right, Outdent Left, Outdent Right, and a 100% zoom level. The text area below the toolbar is empty.

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Question No : 25 of 26

Marks: 5 (Budgeted Time 10 Min) 

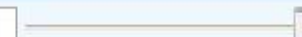
342		
103		
211		
440		
332		
441		
...		

Answer (Please [click here](#) to Add Answer) 

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Time Left

25 

Question No : 25 of 26

Marks: 5 (Budgeted Time 10 Min)

200		
210		
402		
312		
124		

Answer (Please click here to Add Answer)

Rich text editor toolbar with icons for file operations, text formatting, and a 100% zoom level.

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Question No : 25 of 26

Marks: 5 (Budgeted Time 10 Min)

332		
441		
200		
210		
402		
312		

Answer (Please [click here](#) to Add Answer)

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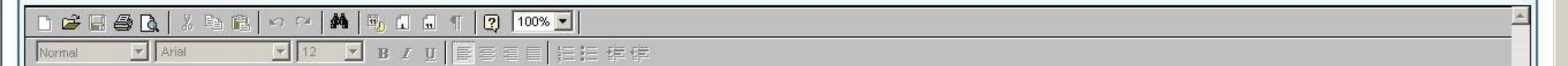


Question No : 26 of 26

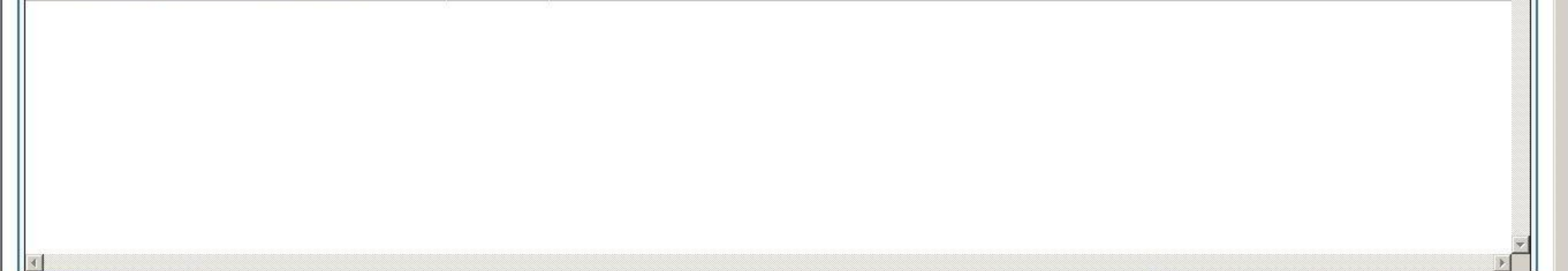
Marks: 5 (Budgeted Time 10 Min) 

Complete the following instance of the optimal matrix multiplication ordering problem,

A1	A2	A3	A4	A5	A6
(6x5)	(5x2)	(2x8)	(8x4)	(4x3)	(3x9)

Answer (Please [click here](#) to Add Answer) 

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Question No : 26 of 26

Marks: 5 (Budgeted Time 10 Min) 

0	1	2	2	2	?
	0	2	2	2	2
		0	3	4	5
			0	4	5
				0	5
					0

Answer (Please [click here](#) to Add Answer) 

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Question No : 26 of 26

Marks: 5 (Budgeted Time 10 Min)

0	60	156	172	184	?
	0	80	104	118	232
		0	64	88	142
			0	96	312
				0	108
					0

Answer (Please [click here](#) to Add Answer)

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59:00

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